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SECTION 01

RETAIL SALES AND INVENTORY MANAGEMENT ANALYSIS

Final Report

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1.0 Introduction

Maintaining huge volumes of data is a must in today's eCommerce world. This project is centered on Olist, a primary marketplace in Brazil that links small businesses with the large retail channels. Olist works with a variety of sellers, products and locations, which means that they process an enormous quantity of data each day.

1.1 Problem Background

Business Intelligence (BI) plays an important role in helping organizations analyze large amounts of data and transform it into meaningful insights for better decision-making. In the retail industry, businesses generate data from various sources such as sales transactions, inventory records, and supplier information. However, these datasets are often stored separately, making it difficult for companies to gain a complete overview of their operations.

- **Business Context**

The e-Commerce industry is known for its wide and diverse products and locations and a significant volume of transactions made. Olist, one of the leading Brazilian department markets that specialized in bridging the gap between small businesses to large businesses. However, there is a challenge in getting the synergy between seller locations, product categories and delivery logistics. The information on sales velocity, logistics cost and regional demand is currently scattered in isolated data sets, and it is challenging for stakeholders to get a holistic picture of the business ecosystem.

- **Importance of the Problem**

For retail organizations, poor inventory management, especially stockouts and overstocking, can result in serious operational and financial issues. Stockouts occur when products are unavailable when customers want to purchase them, resulting in missed sales opportunities and lower customer satisfaction. Overstocking occurs when businesses keep excessive inventory that exceeds customer demand, resulting in operational waste and increase in storage costs. In addition, delays in supplier deliveries may negatively impact inventory planning and product availability. As a result, businesses may struggle to maintain between meeting customer demand and reducing inventory-related costs.

- **Expected Impact**

The expected impact of this project is to create a clear map of the Olist's fragmented data to develop a better understanding. The financial loss from excess stock and missed sales, due to stock cycles, will be avoided by creating an integrated BI solution to ensure that the right products are available for the customer. Rather than trying to piece together data, the stakeholders will be able to visualise the details of delivery and logistics cost on interactive dashboards to see where it is taking place and how it is eating into profits. This project will help the company go from guess to know by leveraging an automated Alteryx workflow to offer quick, accurate insights that can help enhance seller cooperation and general customer satisfaction.

1.2 Objectives

The objectives for the implementation of this project is as follows:

- To integrate disparate datasets (Sales, Products, and Suppliers) into a unified data warehouse using Alteryx Designer.
- To analyze sales performance, inventory trends, and supplier operations in order to identify patterns, inefficiencies, and potential business risks.
- To develop interactive data visualizations and dashboards that provide meaningful insights into retail operations and support data-driven decision-making.
- To enhance inventory planning and retail performance by reducing stock-related issues and improving overall business efficiency.

1.3 Datasets Description

The Brazilian E-Commerce Public Dataset by Olist is utilized as it provides multiple interconnected tables. The dataset was obtained from the open source Kaggle platform. Table 1 provides an overview of the four datasets used.

Table 1: Overview of Brazilian E-Commerce Order Item Dataset

Dataset	Rows	Fields
olist_orders_dataset	99,441	order_id, customer_id, order_status, order_purchase_timestamp, order_approved_at, order_delivered_carrier_date,

		order_delivered_customer_date, order_estimated_delivery_date
olist_order_items_dataset	112,650	order_id, order_item_id, product_id, seller_id, shipping_limit_date, price, freight_value
olist_products_dataset	32,951	product_id, product_category_name, product_name_lenght, product_description_lenght, product_photos_qty, product_weight_g, product_length_cm, product_height_cm, product_width_cm
olist_sellers_dataset	3,095	seller_id, seller_zip_code_prefix, seller_city, seller_state

2.0 Methodology & Workflow

To achieve the project objectives, a structured, data-driven methodology was implemented, focusing on data integration, rigorous transformation and end-user visualization. The workflow is divided into three distinct phases, data acquisition and extraction, ETL (Extract, Transform, Load) Processing and Dashboard design.

Alteryx Designer was selected as the primary ETL tool due to its robust capabilities in handling data sources, data cleansing and complex join operations. The finalized, cleansed data was then staged for presentation in Microsoft Power BI to deliver the interactive and actionable business intelligence.

2.1 Data Workflow Overview

Figure 1 shows the data workflow developed in Alteryx Designer to perform the extraction, cleaning, integration and transformation processes for the datasets obtained. The workflow integrates four separate datasets into a unified master analytical table, which is exported for dashboard development in Microsoft Power BI. The objective of this data workflow is to create an output dataset ready for data visualization on sales performance, inventory trends and supplier operations.

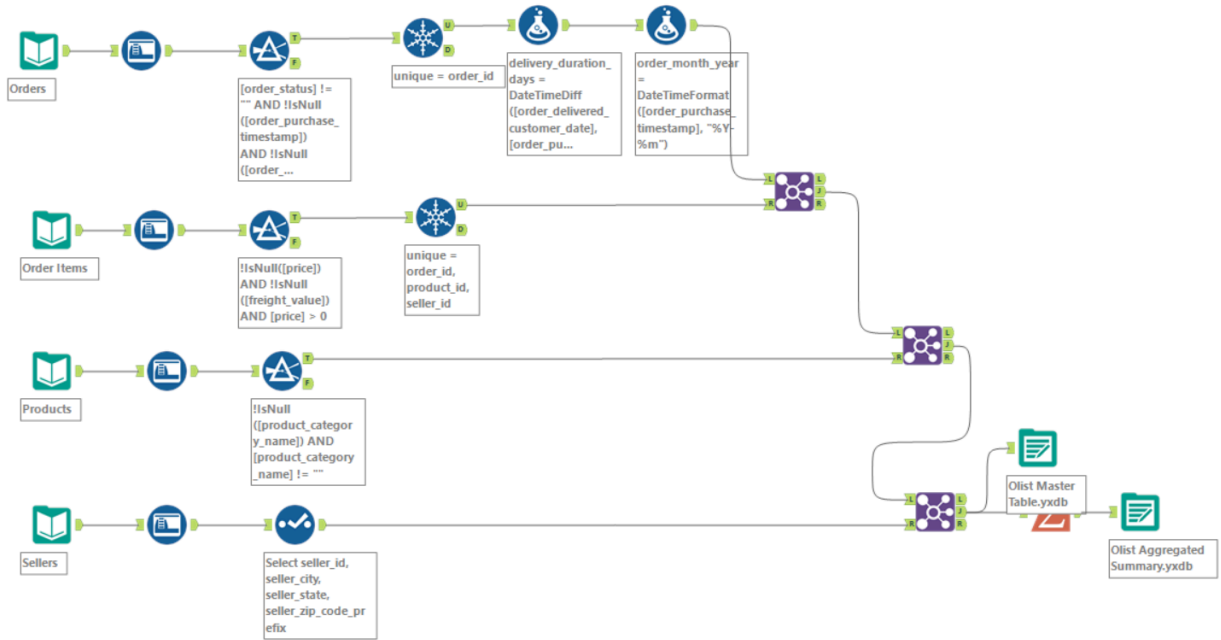


Figure 1: Alteryx Workflow to Create Unified Brazilian Dataset

2.2 Data Input

The input stage begins by importing multiple CSV datasets downloaded from Kaggle into Alteryx Designer using the Input Data Tool. The datasets are loaded directly from local storage directories into the workflow for processing. During import, the input configuration preserves the original dataset formatting and character encoding to ensure Portuguese regional characters and categorical values remain consistent throughout the analytical process.

2.3 Data Cleaning & Data Preparation

Auto Field tool is utilized for each datasets to prevent the attributes to load under generic, memory-heavy V_String types where these nodes automatically optimize allocations down to minimal database structures. Besides, the variable memory allocations for each datasets are compressed into tight, optimized data formats. For instance, the Brazilian E-Commerce Order Item Dataset (olist_order_items_dataset.csv) utilised the Auto Field tool by converting shipping_limit_date field to DateTime and financial fields like price and freight_value to Double as shown in Figure 2.

Before				After			
Record	Name	Type	Size	Record	Name	Type	Size
1	order_id	V_String	254	1	order_id	V_String	254
2	order_item_id	V_String	254	2	order_item_id	V_String	254
3	product_id	V_String	254	3	product_id	V_String	254
4	seller_id	V_String	254	4	seller_id	V_String	254
5	shipping_limit_date	V_String	254	5	shipping_limit_date	DateTime	19
6	price	V_String	254	6	price	Double	8
7	freight_value	V_String	254	7	freight_value	Double	8

Figure 2: Utilising Auto Field for Brazilian E-Commerce Order Item Dataset

The Select tool keeps only the selected field (seller_id, seller_city, seller_state, seller_zip_code_prefix) in order to remove the unnecessary columns to keep the master table lean.

To enforce a strict data consistency and integrity across the item collections for the dataset (Orders streams, Order Items streams & Products streams), a Filter tool is utilised to discards the empty catalog lines (or NULL) where only rows with valid category classifications pass through the True Anchor. For example, *Figure 3* demonstrate the customised filter for Products streams to remove the 610 uncategorised products so category-level analysis in Power BI is accurate.



Figure 3: Customised filter expression for Products Dataset

The Unique tool is applied independently across the primary transaction keys (order_id) in the Order Items stream (olist_order_items_dataset.csv) and Orders stream (olist_orders_dataset.csv). Through the Unique anchor, it extracts the distinct lines by dropping any row duplication to protect metrics against double-counting before the next integration.

2.4 Feature Creation

After the data has been cleaned, Formula tools were used to create 2 new fields which are delivery_duration_days and order_month_year.

The formula used for `delivery_duration_days` is :

***“`delivery_duration_days = DateTimeDiff([order_delivered_customer_date],
[order_purchase_timestamp], "days")`”***

This formula calculates the actual number of days from purchase to delivery, enabling logistics performance analysis. `order_delivered_customer_date` refers to the date the order was delivered to the customer and `order_purchase_timestamp` refers to the date and time the order was placed. `DateTimeDiff` is used to calculate the difference between the two dates.

As for `order_month_year`, the formula used is :

“`order_month_year = DateTimeFormat([order_purchase_timestamp], "%Y-%m")`”

This formula is used to convert the order purchase date and time into a specific text format (e.g., 2018-03). This is done to enable data visualisations to be done using the Line & Cluster Column Chart.

2.5 Data Integration

The workflow connects tables through a 3-stage cascading structure using sequential Join Tools to resolve relational schemas by combining the separate, isolated input vectors into a single, fully enriched master record set without relational data dropouts. In each Join tool's Select tab, the duplicate key column from the right input is deselected to prevent redundant columns in the output. For example the second `order_id` from Order Items.

- **First Join:** Blends the duplicate-free Orders stream (Left) and Order Items stream (Right) via the shared `order_id` field, connecting base orders directly with price logs.
- **Second Join:** Merges the output of the transaction join (Left) with the cleaned Products data stream (Right) using the `product_id` key, mapping category dimensions directly to sales lines.
- **Third Join:** Combining product-order stream (Left) with the Sellers lookup matrix (Right) using the `seller_id` field, appending vital regional attributes like `seller_state` and `seller_city`.

2.6 Data Aggregation

After integration, data aggregation was performed using the Summarise tool. The Summarize Tool is connected to the Join 3 output and aggregates the master table to produce a summary table with only key attributes needed for the data analysis and visualisation. The summary table consists of Seller State, Order Month Year, Product Category Name, Total Revenue, Total Freight Cost, Total Orders and Average Delivery Days fields. Table 2 shows the actions performed on the fields to produce the summary table.

Table 2 : Actions Performed on The Fields to Produce The Summary Table

Action	Field	Output Column Name
Group By	seller_state	Seller State
Group By	order_month_year	Order Month Year
Group By	product_category_name	Product Category Name
Sum	price	Total Revenue
Sum	freight_value	Total Freight Cost
Count	order_id	Total Orders
Average	delivery_duration_days	Average Delivery Days

2.7 Data Output

Once data aggregation was done, the data was exported into csv format using the Output Data tool. Two output files were created. One Output Data tool was connected directly from Join 3 to create an output dataset named “Olist Master Table.csv” which contains all of the data merged into Join 3. Another output dataset was exported named “Olist Aggregated Summary.csv” which was created by connecting the Output Data tool to the Summarise tool. This dataset contains the refined summary table, ideal for data visualisation.

2.8 Tools Used

This section briefly explains about all the tools used in the data workflow along with its purpose of use. Table 3 shows the summary of the tools used.

Table 3 : Summary of Tools Used Along The Data Workflow

Tool	Purpose
Input Data	Load olist_orders, olist_order_items, olist_products, olist_sellers CSVs into the workflow
Auto Field	Convert the data types of fields across all datasets from string to appropriate data type
Filter	Exclude null values in olist_orders, olist_order_items and olist_products
Select	Rearrange the hierarchy of the dataset with seller_id, seller_city, seller_state, seller_zip_code_prefix fields in this stated order
Unique	Remove duplicate entries in olist_orders and olist_order_items dataset
Formula	Create two new fields named delivery_duration_days and order_month_year
Join	Merge all the datasets into one unified dataset
Summarise	Aggregate master table into summarised table with Seller State, Order Month Year, Product Category Name, Total Revenue, Total Freight Cost, Total Orders, Average Delivery Days fields
Output Data	Export two datasets named Olist Master Table and Olist Aggregated Summary into csv files

2.9 Technical Challenges

- Duplicate Join keys across sequential Joins

After Join 1, the output contains two copies of order_id from Orders and Order Items. If not addressed, Join 2 and Join 3 would carry forward the redundant columns, where it can cause ambiguity in the schema. This was resolved by using select configuration inside each Join tool to deselect the duplicate key from the right input before proceeding to the next stage.

- Choosing the right tools either Join or Union tools

The first point of confusion was whether to use the Union tool or Join tool for data integration. The union tool combines tables vertically, while the Join tool combines table horizontally. Since each dataset contains different columns describing aspects of the same

transaction, the Join tool was the best choice. Using Union would have produced a table with many null values.

- **Date Format Parsing for order_month_year**

The order_purchase_timestamp field is stored as a full datetime string. Directly grouping by this field in the Summarize tool would produce one group per unique timestamp. The Formula tool was used to extract only the year-month portion using DateTimeFormat(), producing a clean string like '2018-03' that allows useful monthly aggregation and supports Power BI's visualization.

3.0 Visualisation & Insights

The visualization phase was implemented using Microsoft Power BI Desktop. Three analytical reporting pages were developed using the Olist Master Table and Olist Aggregated Summary datasets generated from the Alteryx workflow which are the Sales Performance Overview dashboard, Logistics & Regional Performance Overview dashboard and the Olist Growth Story 2017 storyboard. These dashboards provide interactive insights into sales performance, logistics efficiency, and regional operational trends, while the storyboard presents the key findings in a narrative format for executive reporting.

3.1 Dashboard 1 : Sales Performance Overview

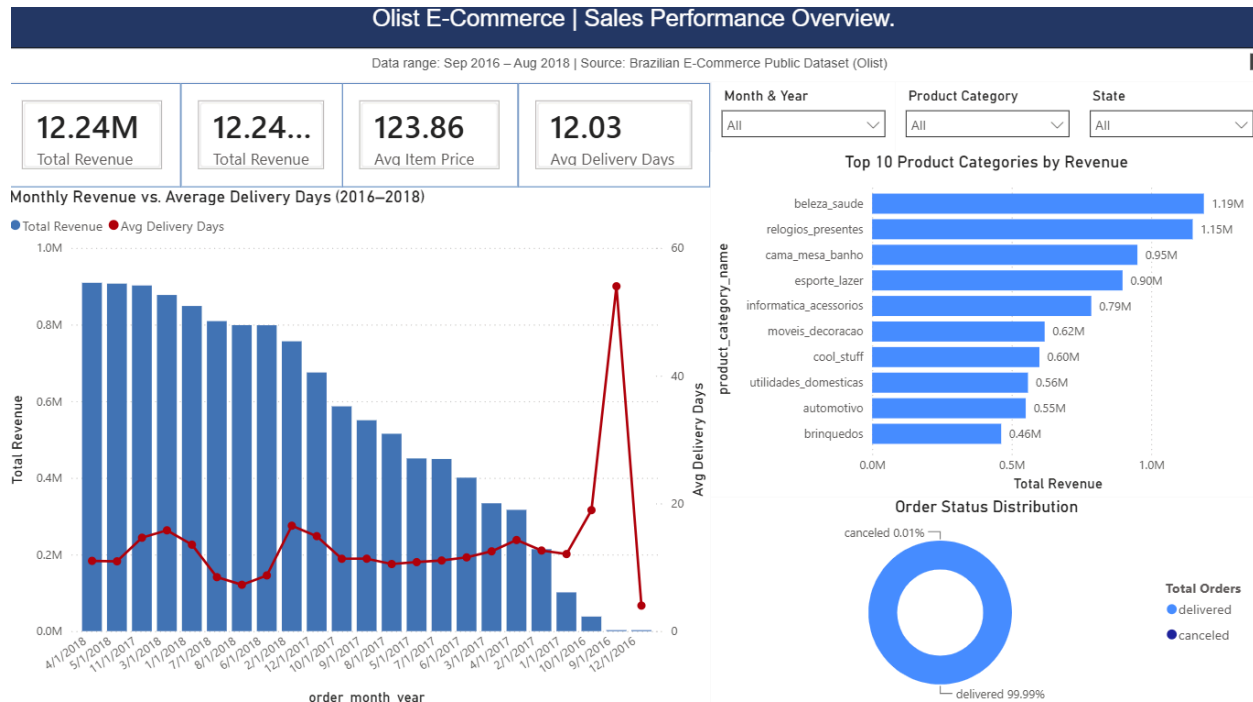


Figure 4: Dashboard 1 - Sales Performance Overview

Dashboard 1 is designed for sales managers and business executives who require a high-level view of business performance. The dashboard combines KPI indicators, trend analysis, product category performance, and order fulfillment status to support strategic decision-making.

Visualization 1 : KPI Cards

Four KPI cards are displayed at the top of the dashboard:

- Total Revenue (R\$12.24M)
- Total Orders (98,666)
- Average Item Price (R\$123.86)
- Average Delivery Days (12.03)

These card visuals provide a summary of key business metrics and allow sales managers and business executives to quickly assess overall business performance. All the KPI cards

dynamically respond to the dashboard slicers, enabling users to view metrics for specific duration, product categories, or seller states.

Visualization 2 : Monthly Revenue vs Average Delivery Days

A combination chart is used to compare monthly revenue trends against average delivery days from September 2016 to August 2018. The blue columns represent Total Revenue while the red line represents Average Delivery Days. The combination chart effectively visualizes two related business metrics with different measurement scales. It enables users to identify whether increases in revenue corresponds with changes in delivery performance, which helps users to evaluate the impact of logistics operations. Users can filter the visualization using the Month & Year, Product Category, and State slicers. Figure 5 shows the dashboard when a specific date is selected, resulting in the dynamic change of Monthly Revenue against Average Delivery Days combination chart.

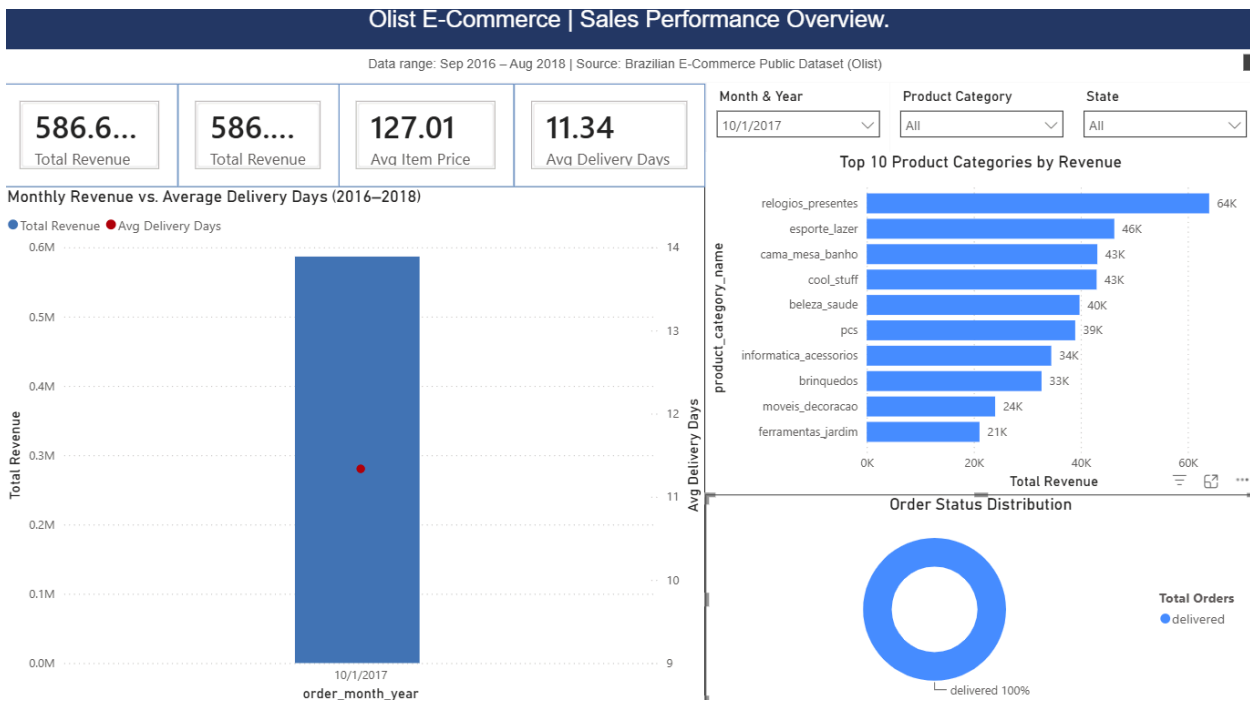


Figure 5 : Dashboard Overview when Specific Date is Selected

Visualization 3 : Top 10 Product Categories by Revenue

A horizontal bar chart was used to rank the ten highest-performing product categories based on Total Revenue. Health & Beauty (beleza_saude) ranks first, followed by Watches & Gifts

(relogios_presentes) and Bed, Bath & Table (cama_mesa_banho). The horizontal bar allows sorting the categories in descending order which helps in identifying the most profitable product segments quickly.

Selecting a category from the horizontal bar chart, such as Health & Beauty (beleza_saude), automatically filters other visuals on the dashboard, allowing users to analyze sales performance and delivery trends for that specific category. Figure 6 shows the interactive dashboard dynamically changing its visualisation tools when Health & Beauty (beleza_saude) is clicked on the Top 10 Product Categories horizontal chart.

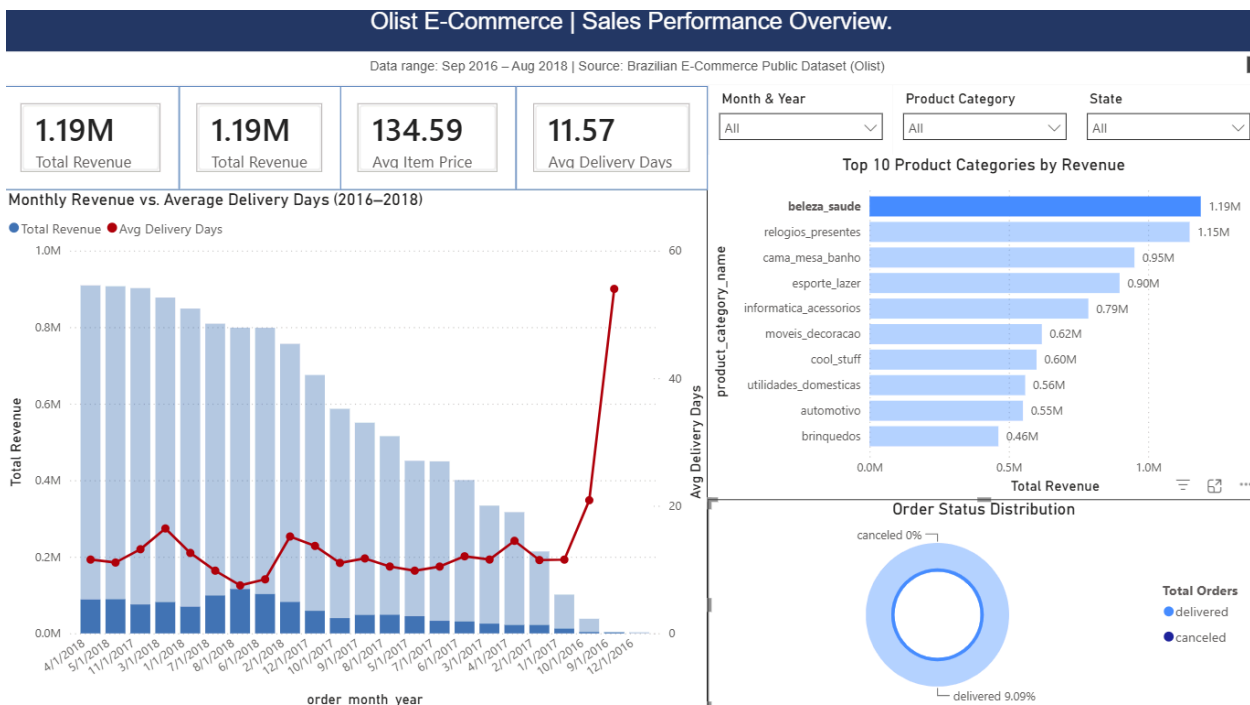


Figure 6 : Dashboard Overview when beleza_saude is Selected on Top 10 Product Categories Horizontal Chart

Visualization 4 – Order Status Distribution

A donut chart was used to display the distribution of order statuses where 99.99% orders were delivered and 0.01% orders were cancelled. Selecting an order status segment filters the other dashboard visuals, enabling further investigation into specific order outcomes.

3.2 Dashboard 2: Logistics & Regional Performance Overview

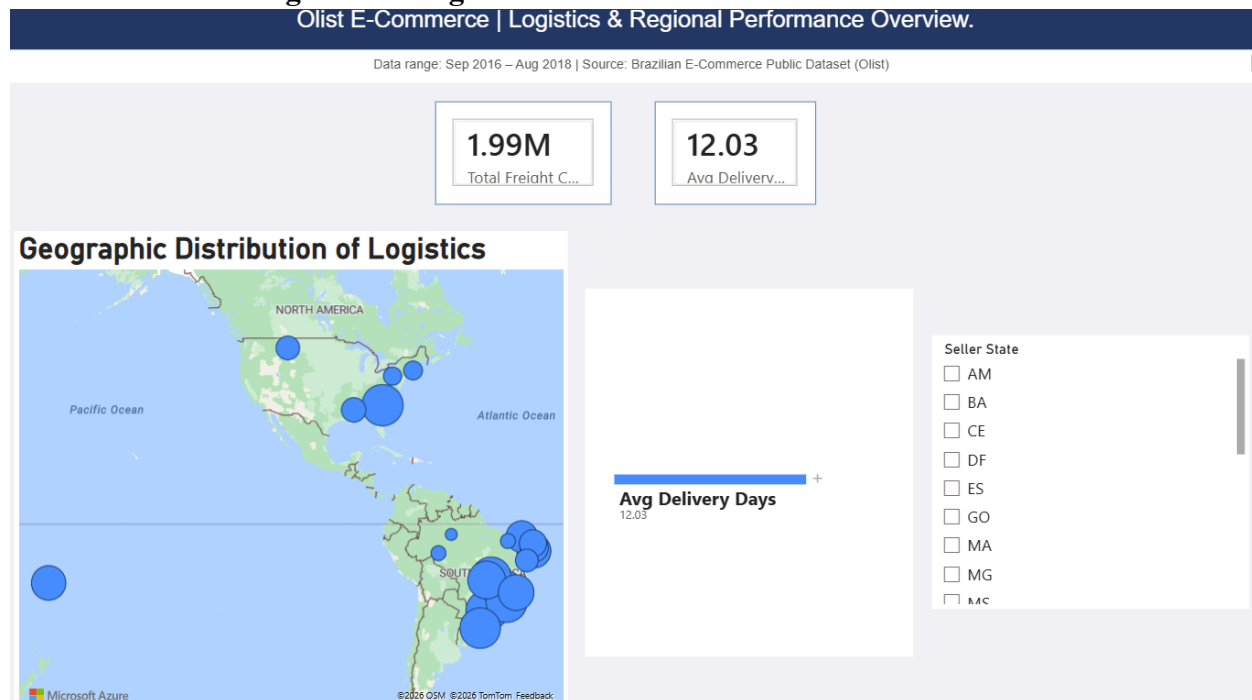


Figure 7 : Dashboard 2 - Logistics & Regional Performance Overview

Dashboard 2 focuses on logistics performance and geographical distribution of freight activities across Brazilian states. Dashboard 2 facilitates logistics managers and supply chain analysts who require visibility into transportation costs and regional delivery efficiency.

Visualization 1 : Logistics KPI Cards

Two KPI cards are displayed at the top of the dashboard which are Total Freight Cost (R\$1.99M) and Average Delivery Days (12.03). These KPI cards provide an overview of logistics performance and allow stakeholders to monitor transportation expenses alongside service efficiency. The KPI values automatically update based on state selections and other applied filters.

Visualization 2 : Geographic Distribution of Logistics

A map was used to visualize logistics activity across Brazil using seller locations. The bubble size in the map represents Total Freight Cost. The larger the bubbles, higher the logistics expenditure. The map shows that logistics activities are concentrated in several major seller states, particularly in the southeastern region of Brazil. Maps were used for geographical analysis

as they allow users to identify regional patterns of logistics activities. The proportional bubble sizes make it easy to compare freight costs across locations.

Visualization 3 : Average Delivery Days Analysis

A bar indicator visual displays the overall average delivery duration of 12.03 days. This visualization provides a benchmark for logistics performance and serves as a reference point when comparing delivery efficiency across different regions. The metric updates dynamically when users filter the dashboard by seller state.

Visualization 4 : Seller State Filter

A Seller State slicer is used in the dashboard to allow users to filter logistics metrics based on specific states. Regional filtering based on the states supports localized performance analysis and allows supply chain managers to investigate freight costs and delivery performance for individual states. The selections made in the slicer update all visuals on the dashboard simultaneously. Figure 8 shows the interactive dashboard dynamically changing when GO state is selected.

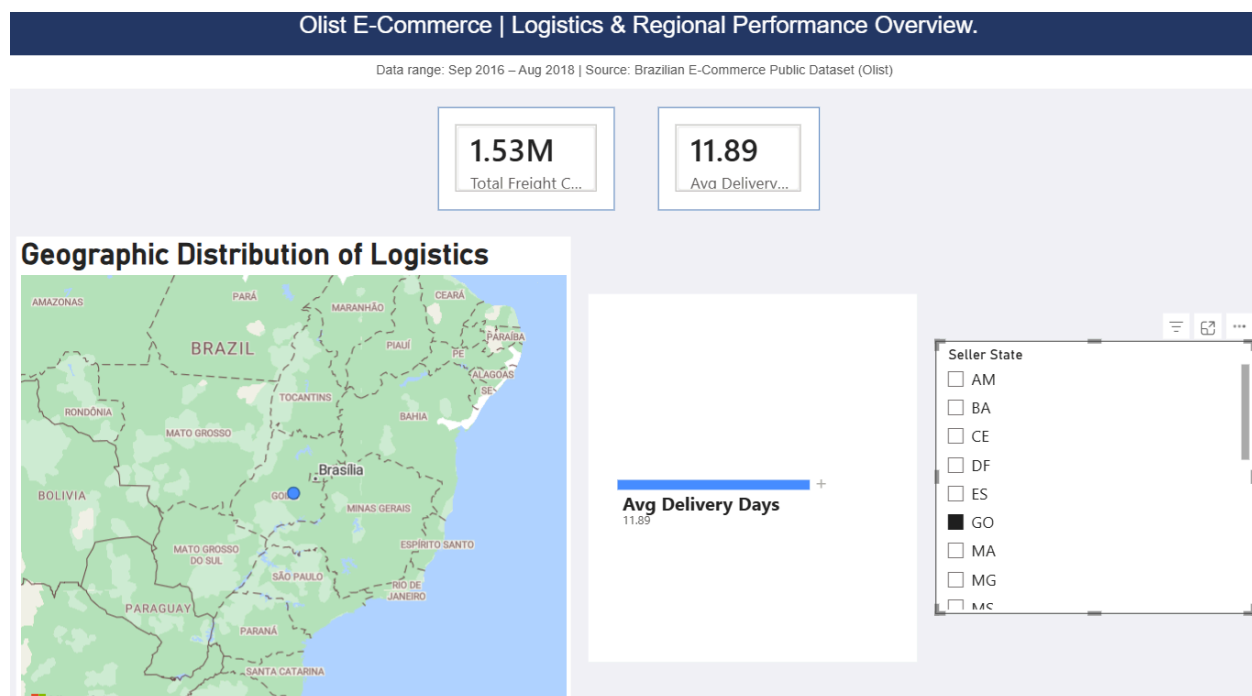


Figure 8 : Dashboard Overview when GO State is Selected

3.3 Storyboard: The Olist Growth Story 2017

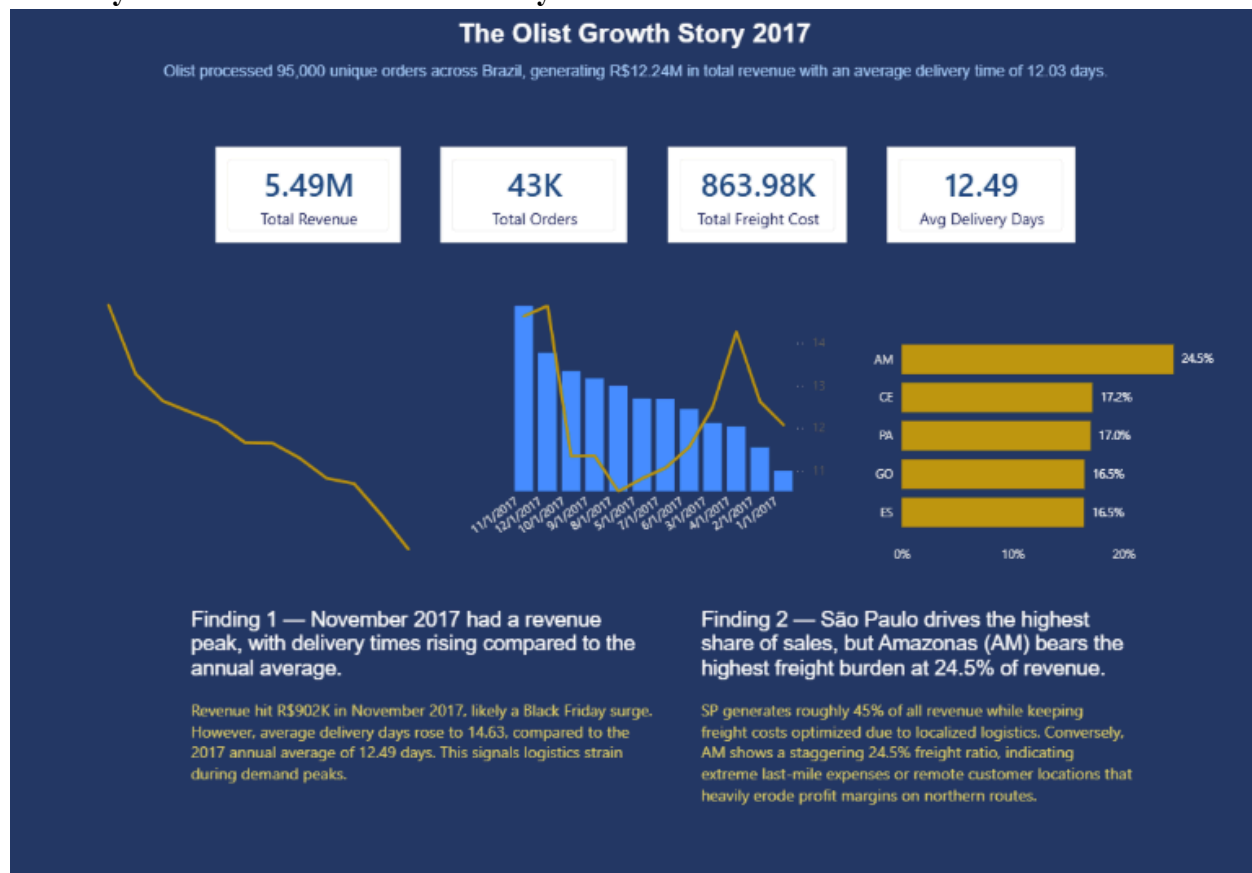


Figure 9 : Storyboard for The Olist Growth Story 2017

The storyboard page presents the key business findings in a narrative format. The storyboard focuses on transforming analytical results into a structured business narrative through a combination of KPIs, charts, and text. This enables non-technical stakeholders and executives to understand the important insights quickly.

The storyboard highlights the company's growth performance during 2017 and provides context behind major business trends.

Finding 1 : Revenue Growth and Delivery Performance

The monthly analysis indicates that November 2017 recorded the highest revenue performance, generating approximately R\$902K in sales revenue. However, average delivery days also increased during this period, reaching 14.63 days compared to the annual average of 12.49 days.

This suggests that during high demand, logistics operations experience more workload which resulted in slower delivery performance.

Finding 2 – Regional Freight Cost Burden

The regional analysis reveals differences in freight cost efficiency across Brazilian states. While major commercial regions such as São Paulo generate revenue, states such as Amazonas experience a much higher freight-cost-to-revenue ratio due to geographical distance and more complex transportation routes. These findings indicate opportunities for logistics optimization and regional distribution improvements.

4.0 Discussion

The analysis of 112,650 order records in the span of September 2016 to August 2018 shows a pattern that will have implications for Olist's operational and strategic decisions.

4.1 Key Insight

The insights for both dashboards and storyboards were derived from analysis of 112,650 order records in the span of September 2016 to August 2018 is explained as in the following:

Sales Performance Insights

- a. The revenue in 2018 reached R\$6.70M in just 8 months, where it surpassed 2017 that reached R\$5.49M. This shows an increase compared to the previous year, which aligns with the swift growth of the marketplace at Olist.
- b. Health & Beauty (beleza_saude) and Watches & Gifts (relogios_presentes) each have a total revenue of R\$1.19M and R\$1.15M respectively make up 19.12% of the total revenue. Especially Health & Beauty products that have the highest average item price at R\$134.59 that shows a high value priority category.
- c. 99.99% of the products with 95,138 orders were successfully delivered, only 0.01% of the orders were cancelled. This indicates a high reliable fulfillment process and a strong accountability on OList platform.

Logistic & Supplier Insights

- a. The map visualization shows that São Paulo (SP) contributes the largest share of total freight costs, costing R\$1.99M followed by states such as Minas Gerais (MG) with total freight costs R\$1.97M and Rio de Janeiro (RJ) with total freight costs R\$1.93M. This indicates that logistics operations are heavily centralized in these regions, resulting in higher transportation expenses and greater dependence on a few major seller hubs.
- b. The dashboard reports an average delivery duration of approximately 12 days. While deliveries are being completed successfully, there is potential to improve logistics efficiency and reduce delivery times to enhance customer satisfaction.

Olist Growth Story of 2017

- a. Peak Demand Logistics Strain (Finding 1): During November 2017, on Black Friday, Peak Demand Logistics's revenues reached R\$902K. This was a high order volume that put a strain on the logistics network, resulting in a rise in average delivery times to 14.63 days, up from the annual average of 12.49 days.
- b. Regional Freight Burden (Finding 2): São Paulo generates the largest proportion of sales, while having low and optimized freight costs. On the other hand, Amazonas (AM) has the largest freight burden with shipping costs accounting for 24.5% of its revenue, reflecting its significant last mile inefficiency on long-distance transportation.

4.2 Business Recommendation

Based on the insights derived from the Sales Performance Overview Dashboard and the Logistics & Regional Performance Dashboard, several data-driven recommendations are proposed to improve business performance and operational efficiency.

Sales Performance Dashboard Recommendations

- a. Since Health & Beauty (beleza_saude) and Watches & Gifts (relogios_presentes) both generate 19.12% of all revenue, these products are categorized in an important category and should be given priority. This means that sellers of these categories need to be kept at above a certain stock level.

- b. Aim to limit the reliance only on top 10 product categories, avoid having too many items in the top 10 categories. If there is a decline in demand in any category, there will be a drop in the overall revenue.

Logistic & Supplier Insights Dashboard Recommendations

- a. Creating additional fulfillment hubs in strategic regions can reduce shipping distances, lower freight costs, and improve delivery speed for customers located far from major seller states.
- b. Using historical sales and logistics data, the company can better forecast demand and allocate resources during peak periods, reducing the risk of delivery delays.
- c. States with the highest freight costs should be continuously monitored through the dashboard to identify inefficiencies and support data-driven decisions for logistics improvement.

Olist Growth Story of 2017

- a. Employ the seasonal capacity planning by carrying out flexible carrier contracts and temporary sorting hubs prior to major sales peaks, such as November, to avoid delivery delays.
- b. Optimize frontier route shipping rates. In certain states, such as AM and CE, freight ratios are high, so it should consider changing the shipping rates to remote areas and negotiating volume discounts with regional carriers when freight ratios are above 17%.

5.0 Conclusion

This project has successfully developed and achieved the Business Intelligence (BI) solution for Olist's Brazilian e-Commerce operations by fulfilling all four project objectives during the three project phases. In Phase 1, the problem background, business context and the structure of the dataset was established by identifying the key challenges of the fragmented data, inventory management risk and lack of efficiency in logistics as areas where the BI can be applied. The problem identified led to the creation of the data workflow using Alteryx Designer by integrating the four separate CSV files dataset that consists of orders, order items, products and sellers to produce one master table and an aggregated summary that is ready to be used in the next phase.

This project is continuing to Phase 3 in designing two interactive dashboards and 1 storyboard that were developed using Microsoft Power BI. Sales Performance Overview dashboard showed the growth in revenue by year whereas Health & Beauty (beleza_saude) and Watches & Gifts (relogios_presentes) both are the two most profitable revenue with a high demand logistic performance. The Logistic & Regional Overview dashboard shows that freight cost distribution was plotted geographically by state in Brazil. The storyboard compiled these results into a format that was suitable for executive presentation as a story.

The BI solution allows Olist's stakeholder to shift from data guess work to making decisions from the driven data through the concrete business recommendation that derived directly from the dashboard insights. Future enhancement that can be improved is by including the real-time data integration through Microsoft Power BI that allows customer review sentiment analysis and predictive delivery day modeling by using machine learning to further enrich and strengthen the BI ecosystem.

6.0 Reference

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